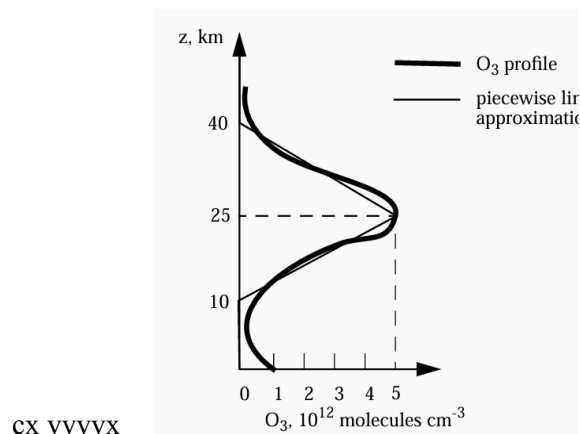


Assignment 1: EES 202 Introduction to atmospheric and climate sciences

1. A weather station reports $T = 293 \text{ K}$, $\text{RH} = 50 \%$ at sunset. Assuming that $P_{\text{H}_2\text{O}}$ remains constant, by how much must the temperature drop over the course of the night in order for the fog to form?
2. What is the mass concentration of water vapor ($\text{g H}_2\text{O}$ per m^3 of air) in a liquid-water cloud at a temperature of 273 K ? Considering that the liquid water mass concentration in a cloud ranges typically from 0.1 to 1 g liquid water per m^3 of air, is most of the water in a cloud present as vapor or as liquid?
3. Consider the following typical vertical profile of ozone (O_3) number densities measured over the United States. Ozone is produced in the stratosphere ($10\text{-}50 \text{ km}$ altitude) by photolysis of O_2 and subsequent combination of O atoms with O_2 . The stratospheric O_3 layer protects life on Earth by absorbing solar UV radiation and preventing this radiation from reaching the Earth's surface. Fortunately, the O_3 layer is not in contact with the Earth's surface; inhalation of O_3 is toxic to humans and plants, and the U.S. Environmental Protection Agency (EPA) presently has an air quality standard of 80 ppbv O_3 not to be exceeded in surface air.



- 3.1 Calculate the mixing ratio of O_3 at the peak of the O_3 layer ($z = 25 \text{ km}$; $P = 35 \text{ hPa}$; $T = 220 \text{ K}$). Would this mixing ratio be in violation of the EPA air quality standard if it were found in surface air?
- 3.2 Calculate the mixing ratio of O_3 in surface air ($z = 0 \text{ km}$; $P = 1000 \text{ hPa}$; $T = 300 \text{ K}$). Is it in compliance with the EPA air quality standard? Notice that the relative decrease in mixing ratio between 25 km and the surface is considerably larger than the relative decrease in number density. Why this?
- 3.3 The total number of O_3 molecules per unit area of Earth's surface is called the O_3 column and determines the efficiency with which the O_3 layer prevents solar UV radiation from reaching the Earth's surface. Estimate the O_3 column in the above profile by approximating the profile with the piecewise linear function shown as the thin solid line.

- 3.4 To illustrate how thin this stratospheric O₃ layer actually is, imagine that all of the O₃ in the atmospheric column was brought to sea level as a layer of pure O₃ gas under standard conditions of temperature and pressure (1.013×10^5 Pa, 273 K). Calculate the thickness of this layer.
4. Oxygen has a fixed mixing ratio in the atmosphere. How would you expect its number density measured in surface air to vary between day and night? How would you expect its partial pressure measured in surface air to vary between day and night?
5. Give a rough order of magnitude for the number of molecules present in a typical 1-micrometer aerosol particle.
6. Consider an air parcel at sea level ($P = 1013$ hPa) with 30 °C and 50 % RH. The saturation water vapor pressure at 30 °C is 42 hPa. Show that the water mixing ratio in this air parcel is $C_{\text{H}_2\text{O}} = 0.021$ mol mol⁻¹.
7. Assume a CO₂ mixing ratio of 410 ppm in that sea-level air parcel. As the air parcel rises through the air, its water vapor may be removed by cloud formation and precipitation. Assuming 100 % removal of water and no dissolution of CO₂ in the condensed water, what is the mixing ratio of CO₂ in the dried-out air parcel?